Pendles On Sui - Why not?

Presenter: Link


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(Apple pen !)
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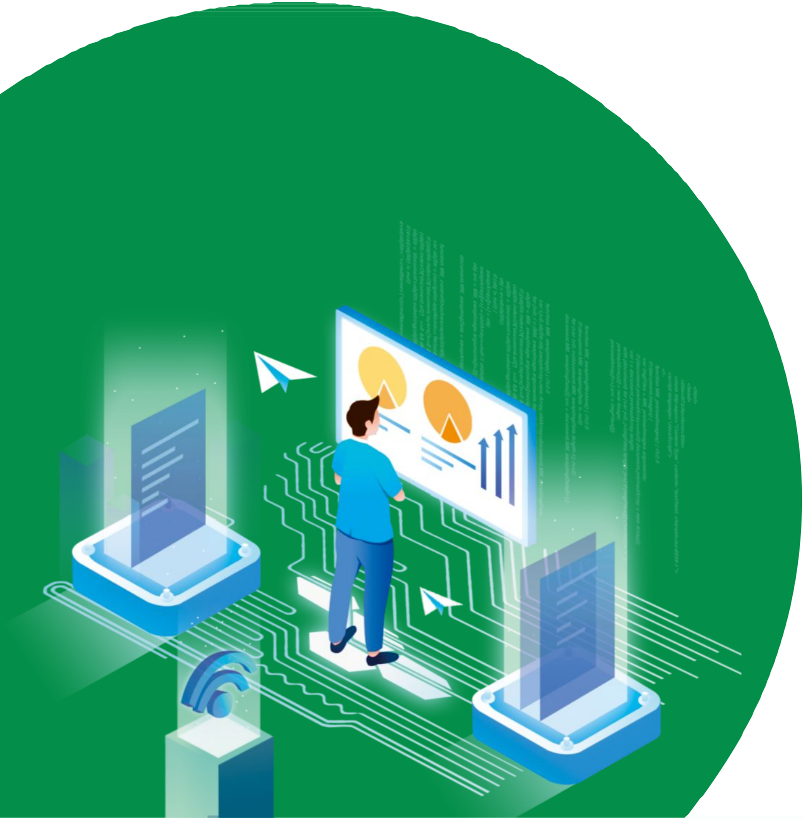


CONTENT

The Foundation: Yield-Bearing Assets

Yield-Bearing Assets (YBAs): Assets that automatically accumulate yield over time

Traditional Finance

- Earning interest (Bank)
- Coupon payments (VIC,...)

DeFi

- LP tokens -> earn trading fee,...
- LS tokens (stSUI) -> earn staking rewards
- LR tokens -> earn re-staking rewards
- Lending tokens (nSUI -Navi, sSUI - Scallop) -> earn -> earning interest

Yield-Bearing Assets in DeFi Problems

Raw Yield Assets:

- Have an exchange rate vs underlying token
- Exchange rate changes over time due to yield
- Yield is variable and depends on market conditions
- Yield is bundled: cannot be traded separately

The Market Need:

- A demand for predictability led to the emergence of fixed-yield instruments
- Users can trade future yield

Pendles Solution

Solution: Pendle introduces a system to split the asset:

- **Principal (PT)** -> underlying value
- **Yield (YT)** -> right to future yield

This enables:

- Fixed yield investing
- Yield speculation
- Market-driven yield pricing
- New AMM mechanics

Pendle Finance Core: Tokenizing Yield

The Three-Step Mechanism:

- **Wrap**: A yield-bearing asset (e.g., nSUI) is deposited and wrapped into a Standardized Yield Token (SY)
- **Split**: The SY is split into two separate tokens:
  - Principal Token (PT): Represents the underlying principal amount. Its value appreciates to par at maturity.
  - Yield Token (YT): Represents the right to all future yield. Its value decays to zero at expiry.
- **Settle**: At maturity, PT is redeemable 1:1 for the underlying asset, and YT expires.

The Magic of Pricing & User Motivation

Core Pricing Identity:

`Price(Underlying Asset) = Price(PT) + Price(YT)`

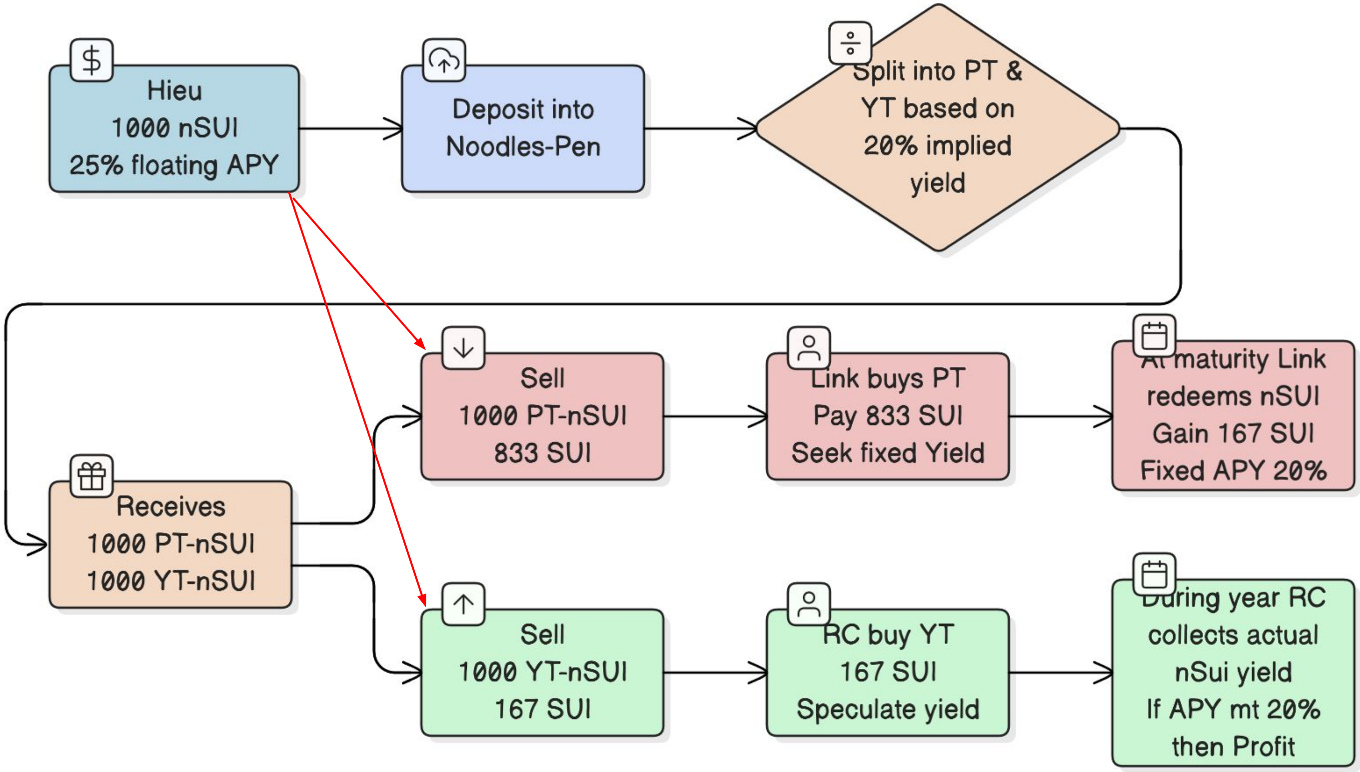
Why Buy PT? (The Fixed-Yield Seeker):

- Goal: Lock in a known, fixed return.
- Example: Buy PT-nSUI at a discount (e.g., 0.8 SUI). At maturity, redeem it for 1 nSUI (worth 1 SUI + accrued yield). The discount equals your fixed yield.

Why Buy YT? (The Yield Speculator):

- Goal: Bet on future yield performance.
- Example: Buy YT-nSUI for 0.2 SUI. You receive all yield generated by the underlying nSUI. Profit if actual yield > market's implied yield when you bought

Pendle Finance Core: Tokenizing Yield



Pendle's AMM: Trading Time-Decaying Assets

The Challenge:

- PT and YT are time-decaying assets.
- A standard Constant Product (*x*y=k*) AMM is inefficient.

Pendle's Solution: AMM curve includes a *time factor* that continuously modifies pricing.

Key Insight: The AMM curve dynamically adjusts based on time to maturity

- Far from maturity: More liquidity, lower slippage for establishing rates.
- Near maturity: Liquidity concentrates, reflecting PT's convergence to par and YT's decay.
- Price:
  - ***PT Price***: Starts discounted, converges to 1.0 SUI at maturity
  - ***YT Price***: Starts at premium for yield rights, decays to 0
  - Sum Always Equals Underlying: *PT Price + YT Price = SY* (price of nSUI)

Noodles-Pen! Why not?

User Type	Current Sui DeFi	With Noodles-Pen	Advantage
Conservative Investor	Accept variable yield (3-15% APY)	Buy PT for fixed yield (e.g., 8% APY)	Predictability
Yield Farmer	Chase highest APY, impermanent loss	Buy YT to speculate on yield movements	Pure yield exposure, no IL
Liquidity Provider	Provide SUI/nSUI to AMM, earn fees	Provide PT/YT liquidity, earn fees + incentives	Higher fees, less volatility
Protocol Treasury	Hold assets, yield fluctuates	Sell YT for upfront cash, hedge risks	Immediate liquidity, budgeting

Noodles-Pen! Why not?

Suitability Analysis:

- **YBA:** Scallop, Navi (Lending), Aftermath, Cetus (LST/LPs).
- **Move contract**: SY Standards, Minting/Burning of PT & YT,...
- **Object-based model vs Account-based model**: <RC will tell you>

Noodles-Pen! Why not?

Todo list:

- **Research & Detailed Design:** Account-based model -> Object-based model, Move models, simplified AMM math,..
- **Core Contracts:** SY, Minting, Vault,...
- **Custom AMM**

References

- **Pendles Finance v2 Docs:** **https://docs.pendle.finance/pendle-v2/ProtocolMechanics/Glossary (Highly recommended)**
- **Pendles Finance V2 source:** **https://github.com/pendle-finance/pendle-core-v2-public/tree/main (Deep and dive)**